

LIGHTWEIGHT SAFE HELICOPTER (LSH)

GENERATION 1 — V4.0 ENGINEERING DOCUMENT TREE

Controlled Documentation Architecture for Engineering Development, Licensee Due Diligence and Investment Review

Document Purpose

This document defines a practical engineering-document hierarchy for LSH V4.0 Generation 1. It is intentionally structured as a controlled engineering development tree rather than a 500-document presentation package. The tree is designed to support requirements allocation, detailed design, simulation, safety analysis, subsystem development, prototype integration, verification and evidence generation, while also allowing a buyer/licensee or investor to review the program through a controlled subset.

Baseline used: LSH V4.0 Document 00, which defines a 20,000 kg target, 21.0 m main rotor, 3.5 m tail rotor, 1,000 km range target, occupied survival capsule, eight fixed supports, selective separation, deterministic Safety Kernel and staged V&V; gates. These remain targets/concepts until engineering evidence closes them.

1. Recommended Document Count

Documentation level	Recommended count	Purpose
Core controlled baseline / program architecture	20–30	System definition, requirements, architecture, interfaces, configuration, safety basis.
Engineering development tree	~280–360	Detailed subsystem specifications, analyses, drawings/data definitions, software, structures, propulsion, recovery and integration.
Full evidence / qualification repository	~400–550+	Detailed analysis reports, test procedures, test reports, inspection records, qualification evidence, configuration records and certification artifacts.
Licensee / investor controlled package	~35–70	Decision-ready subset showing architecture, requirements maturity, risk, evidence, roadmap, IP surfaces and commercial path without exposing the entire engineering repository.

Conclusion: A 500-document tree is technically reasonable as a *full controlled engineering/evidence repository*, but it is too large to be the default document set handed to an investor or license buyer. The recommended strategy is one master tree plus a smaller controlled due-diligence subset.

2. Master Engineering Document Tree

00–019 — Program, System Baseline & Requirements

Document 00; system requirements; configuration management; mission definition; architecture; design philosophy; terminology; requirements allocation; compliance basis.

020–059 — Safety, Hazard & Survival Architecture

System safety; FHA/PHA; emergency states; Safety Kernel; independence/common-cause; failure modes; hazard logs; survival cases; human authority.

060–099 — Survival Capsule & Occupant Protection

Capsule requirements; protective cell; internal roll-cage/survival cell; seats/restraints; crash loads; energy management; egress; fire/smoke; post-landing survival.

100–139 — Rapid Selective Separation & Jettison

Separation requirements; release mechanisms; locks/inhibits; command paths; fuel isolation; release-state sensing; partial/asymmetric release; trajectory; clearance; ground tests.

140–169 — Parachute, Recovery & Descent

Canopy sizing; deployment; reefing; risers; failure tolerance; stabilization; inflation dynamics; descent control; recovery avionics; optional residual-airframe recovery trade.

170–219 — Airframe, Structural & Landing Supports

Primary structure; capsule interfaces; load paths; fatigue; damage tolerance; crashworthiness; 8 support geometry; energy absorbers; terrain/slope/soft-ground cases.

220–259 — Rotor, Transmission & Propulsion

Main rotor; tail rotor; hub; controls; transmission; gearboxes; engines; mounts; cooling; fuel; OEI; autorotation; vibration; thermal margins.

260–299 — Avionics, Electrical, Controls & Software

Flight controls; emergency controls; Safety Kernel implementation; sensors; power; independent emergency power; data buses; software; health monitoring; AI boundary; cybersecurity.

300–339 — Aerodynamics, Performance & Mission

Aerodynamic model; hover; climb; cruise; drag; hot/high; OEI; autorotation; range; fuel; payload; environmental envelope; performance maps.

340–379 — Dynamics, 6-DOF & Emergency Simulation

Aircraft 6-DOF; capsule 6-DOF; released components; asymmetric release; rotor states; parachute inflation; terrain interaction; Monte Carlo/sensitivity; model correlation.

380–429 — Interfaces, Integration & Manufacturing Definition

ICDs; harnesses; hydraulics; fuel interfaces; mechanical interfaces; assembly; maintainability; tooling; materials/processes; manufacturing readiness.

430–479 — Verification, Validation, Test & Qualification

V&V; matrix; analysis plans; simulation verification; subsystem tests; integrated ground tests; crash tests; parachute tests; flight test plans; qualification evidence.

480–519 — Certification, Configuration, Quality & IP Evidence

Certification basis; compliance evidence; quality records; configuration index; change control; requirements traceability; IP extraction; prior-art/legal work products; controlled release packages.

3. Proposed Document-ID Architecture

Prefix	Branch	Examples
LSH-00-xxx	System baseline	LSH-00-001 Master System Description; LSH-00-010 System Requirements
LSH-02-xxx	Safety	LSH-02-020 Safety Architecture; LSH-02-040 Safety Kernel Requirements
LSH-06-xxx	Capsule	LSH-06-001 Capsule Requirements; LSH-06-030 Protective Cell
LSH-10-xxx	Separation	LSH-10-001 Separation System Requirements; LSH-10-050 Release Mechanism
LSH-14-xxx	Recovery	LSH-14-001 Parachute Requirements; LSH-14-040 Deployment Analysis
LSH-17-xxx	Structure/Landing	LSH-17-001 Structural Basis; LSH-17-060 Landing Support & Energy Absorption
LSH-22-xxx	Propulsion/Rotors	LSH-22-001 Propulsion Requirements; LSH-22-040 Main Rotor
LSH-26-xxx	Avionics/Software	LSH-26-001 Avionics Architecture; LSH-26-040 Safety Kernel Software
LSH-30-xxx	Performance	LSH-30-001 Performance Requirements; LSH-30-050 Range Analysis
LSH-34-xxx	Dynamics	LSH-34-001 6-DOF Model Requirements; LSH-34-060 Separation Dynamics
LSH-43-xxx	V&V/Test	LSH-43-001 Master V&V; Plan; LSH-43-050 Integrated Ground Test
LSH-48-xxx	Certification/IP	LSH-48-001 Certification Basis; LSH-48-050 IP Evidence Register

The exact document IDs should be assigned only when the document is created or formally allocated. Do not fabricate traceability merely to make the tree appear complete.

4. Buyer / Licensee Due-Diligence Package

The buyer should not receive 500 uncontrolled PDFs at the first stage. A professional package should be a controlled subset with a document index and evidence status. A practical first disclosure package is approximately 45–60 documents:

01. Document 00 — Comprehensive Engineering Baseline
02. System Requirements + Architecture
03. Mission / Operational Concept
04. System Safety Architecture + preliminary hazard analysis
05. Survival Capsule specification + protective-cell concept
06. Rapid Selective Separation specification
07. Safety Kernel and emergency decision architecture
08. Recovery / parachute architecture
09. 8-support landing and energy-absorption specification
10. Mass / CG / inertia baseline
11. Rotor / propulsion preliminary sizing
12. 1,000 km mission and range model
13. 6-DOF emergency dynamics plan/model definition
14. Interface Control Document set
15. Software / controls / autonomy boundary
16. V&V; Master Plan + evidence matrix
17. Technology readiness / development gate plan
18. Risk register + critical open items
19. IP element register + invention disclosures
20. Prior-art / patent work products when available
21. Commercialization / prototype roadmap
22. Configuration index and document status register

Disclosure principle: the investor/licensee package should demonstrate that the architecture can be engineered and that the open items are explicitly controlled. It should not pretend that conceptual targets are already verified. Document 00 itself states that the V4.0 baseline is preliminary and that mass, rotor/propulsion, 6-DOF, separation, parachute, crash and certification evidence remain development gates.

5. Engineering Maturity Path

Gate	Name	Primary evidence
G0	Architecture Freeze	Controlled baseline, system boundaries, interfaces and emergency states.
G1	Analytical Closure	Mass/CG/inertia, rotor, propulsion, structural load paths, separation kinematics, preliminary mission closure.
G2	Subsystem Demonstration	Release hardware, locks/inhibits, energy absorbers, parachute components, sensors and independent power.
G3	Integrated Ground Demonstration	End-to-end sequence, timing, state sensing and capsule-clear verification.
G4	Dynamic / 6-DOF Simulation	Representative emergency states, partial/asymmetric release and parachute inflation.
G5	Structural / Crash Testing	Capsule, attachment points, seats, landing supports and occupant-protection evidence.
G6	Recovery System Testing	Canopy deployment, inflation loads, failure cases, stability and landing performance.
G7	Flight Development	Controlled flight tests within the approved development envelope.
G8	Certification / Qualification	Formal evidence against the selected certification basis.

Document 00 V4.0 explicitly defines these gates and states that the baseline is suitable for the next engineering phase, but not yet a certified aircraft specification or proof of achieved 20-ton performance, 1,000 km range or survival probability.

6. Final Recommendation

Recommended master tree: retain a 480–520 document controlled architecture for Generation 1, expandable to roughly 550+ when detailed evidence, manufacturing and qualification records are included.

Recommended buyer-facing package: approximately 45–60 core documents at first disclosure, expanding through staged due diligence.

Recommended investor-facing package: approximately 15–25 high-level controlled documents, with evidence summaries and a secure data-room index.

Important: document count is not engineering maturity. A smaller tree with closed requirements, analyses and test evidence is stronger than a 500-document tree filled with placeholders. The purpose of the tree is to provide controlled coverage of every engineering discipline and a clear path from concept → analysis → prototype → test → qualification.

STATUS: [PROPOSED] Controlled Generation 1 documentation architecture. Document counts are planning targets and shall be refined as the engineering work packages are allocated.